Can You Hear the Shape of an Organ? Practical Identifiability of Elastic Shell Parameters from Resonant Frequencies

Jonathan Mace[a,*], Brian R. Mace[b]

[a]*Microsoft Research, Redmond, WA, United States*
[b]*Acoustics Research Centre, Department of Mechanical and Mechatronics Engineering, University of Auckland, Auckland, New Zealand*

Abstract

We investigate the practical identifiability of material and geometric parameters of a fluid-filled elastic oblate spheroidal shell from its resonant frequencies — the inverse counterpart of Kac's celebrated question "Can one hear the shape of a drum?" applied to vibroacoustic organ models. Using an oblate Ritz flexural forward model — with the equivalent-sphere reduction as a degenerate comparator — we compute the scaled Jacobian of the frequency–parameter map, its singular values, its condition number, and the associated Fisher information matrix for the parameter triple (a, c, E). The smallest singular value of the scaled Jacobian admits a *two-channel* regular expansion $\sigma_{\min}(\varepsilon) = \sigma_0 + \lambda_1\varepsilon^2 + O(\varepsilon^4)$, where the *curvature channel* $\sigma_0 > 0$ reflects mode-dependent Legendre weights that break the a/c degeneracy even at the sphere, and the *shape channel* $\lambda_1\varepsilon^2$ captures eccentricity-driven lifting. Within the Ritz model (five modes), the condition number drops from a near-spherical floor $\kappa_{\text{floor}} \approx 269$ to $\kappa = 69.4$ at the canonical oblate operating point, an improvement by a factor of approximately 3.9. For comparison, the equivalent-sphere model gives κ numerically singular ($\kappa > 10^{10}$ in finite-difference evaluation with step $\delta = 10^{-6}|\theta_j|$), reflecting algebraic rank deficiency rather than geometric conditioning. A Newton-type inverse solver recovers (a, c, E) to relative errors below 10^{-12} in round-trip tests, while Cramér–Rao bounds at 1% frequency noise yield uncertainties of ±22.5% in a, ±6.0% in c, and ±30.8% in E (five modes). The two-channel framework leads to a three-class classification of shell inverse problems: *singular* (equivalent-sphere, rank-deficient), *regular bounded* (oblate Ritz, κ approaches a finite floor and generally improves with eccentricity, exhibiting local non-monotonicity from mode crossings), and *non-lifting* (prolate, κ flat and high). The oblate–prolate asymmetry arises from curvature–mode anti-correlation: in oblate shells, peak curvature and modal peaks occupy different latitudes, diversifying the Jacobian columns; in prolate shells, both concentrate at the poles, leaving the columns nearly proportional. One can "hear the shape" of an oblate organ, but not a spherical or prolate one.

Keywords: inverse problem, identifiability, elastic shell, resonant frequency, condition number, Cramér–Rao bound, oblate spheroid, Kac problem

1. Introduction

In 1966, Kac asked the now-classic question "Can one hear the shape of a drum?" [1]. The question is spectral: does the set of resonant frequencies determine the geometry uniquely? Gordon, Webb, and Wolpert [2] later showed

*Corresponding author
Email addresses: jonathanmace@microsoft.com (Jonathan Mace), br.mace@auckland.ac.nz (Brian R. Mace)

that, for planar drums, the answer is "no": distinct domains can share the same Laplacian spectrum. That result did not close the subject. It made it more interesting.

The Browntone programme has approached that question from the forward side. Paper 1 established that the human abdomen supports low-order flexural resonances in the infrasonic band [3]; Paper 4 extended the same fluid–shell framework to the fill-dependent resonance of the urinary bladder [4]; and Paper 7 showed that an analogous model can invert a watermelon tap-tone into a rind modulus estimate [5]. Those papers answer variants of the question "what frequencies does this shell produce?" The present paper asks the inverse question on which all such applications ultimately depend: given measured frequencies, can one recover the shell's material properties at all?

That identifiability question is the theoretical keystone of the programme. Without it, a forward resonance model is descriptive but not diagnostic. With it, the same spectrum can become the basis of non-invasive elastography, parameter estimation, and experimental design. In practical terms, if a small set of resonant frequencies determines stiffness and geometry with acceptable conditioning, then the abdominal, bladder, and watermelon problems cease to be curiosities of shell vibration and become inverse tools. If not, one merely fits a model to a spectrum that never had enough information in the first place.

Here we therefore study an engineering analogue of Kac's inverse problem for a fluid-filled elastic shell. We focus on low-frequency flexural modes $n \geq 2$, rather than the breathing mode $n = 0$, which lies near 2490 Hz and is governed primarily by fluid compressibility. The inverse problem is posed on the measured flexural frequencies $\{f_2, f_3, \ldots, f_N\}$, with the aim of recovering the semi-major axis a, the semi-minor axis c, and the Young's modulus E.

The key geometric point is oblateness. A spherical shell sits at a spectrally degenerate limit: symmetry forces geometric sensitivities to collapse, so the inverse map becomes singular or nearly so. An oblate shell does not enjoy that luxury. Once $a \neq c$, different flexural modes sample curvature differently, the degeneracy is lifted, and identifiability can in principle be restored. The question is not whether asymmetry looks promising in a qualitative sense, but whether it restores enough rank to support useful inversion.

A central modelling point must also be stated plainly. The identifiability analysis uses an *oblate Ritz model* that integrates strain energy over the spheroidal surface, so that each flexural mode samples meridional curvature differently. For comparison, we also examine the equivalent-sphere reduction

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{1}$$

which collapses all curvature information into a single radius. The contrast between these two models illuminates the role of geometry in identifiability: the equivalent-sphere model is algebraically rank deficient by construction, whereas the Ritz model restores full rank for any eccentricity $\varepsilon > 0$.

Our contribution is therefore twofold. First, we quantify practical identifiability through the Jacobian of the frequency–parameter map, using the dimensionless scaled Jacobian, its singular values, its condition number, and the associated Fisher information matrix to assess whether small perturbations in the measured frequencies can be mapped reliably back to (a, c, E). Second, we make explicit the programme-level consequence: identifiability, not merely forward prediction, is the mathematical foundation on which Papers 1–7 rest. This is an analysis of *practical* identifiability in the sense of Raue et al. [6] and Wieland et al. [7], not a global uniqueness theorem.

The main numerical findings are as follows.

- Within the Ritz model, the condition number drops from a near-spherical floor value $\kappa_{\text{floor}} \approx 269$ (five modes) to $\kappa = 69.4$ at the canonical oblate operating point ($\varepsilon = 0.745$), purely through geometric eccentricity.
- The equivalent-sphere model gives $\kappa = 1.37 \times 10^{10}$ (algebraically rank deficient), illustrating the severity of model-induced degeneracy. By Proposition 1, κ is formally infinite; the finite value is an artefact of the finite-difference step $\delta = 10^{-6}|\theta_j|$.
- The smallest singular value admits a regular two-channel expansion $\sigma_{\min} = \sigma_0 + \lambda_1\varepsilon^2 + O(\varepsilon^4)$; the curvature channel dominates everywhere ($\varepsilon_c = 1.48 > 1$).
- At 1% frequency noise (five modes), the Cramér–Rao lower bounds correspond to uncertainties of approximately ±22.5% in a, ±6.0% in c, and ±30.8% in E — comparable to existing elastographic precision.
- The identifiability metrics are robust to mode count: the condition number remains in the range $\kappa \in [69, 94]$ for $\mathcal{N} = 3$ to 9 modes.
- Three-mode inversion is locally feasible but exhibits local minima, whereas five-mode inversion is robust across the tested initial guesses.
- A three-class taxonomy emerges: *singular* (equivalent-sphere, rank-deficient), *regular bounded* (oblate, κ finite and generally favourable, with local non-monotonicity from mode crossings), and *non-lifting* (prolate, κ high and flat). The oblate–prolate asymmetry arises from curvature–mode anti-correlation.

These results suggest that one can, in a practical sense, hear the shape of an oblate organ from a modest set of flexural resonances, whereas the same task is poorly posed at the spherical limit. For the Browntone programme, that distinction is not a side remark. It explains why the forward models of Papers 1–7 are potentially invertible in the first place, and why oblateness rather than near-sphericity is the feature that turns resonance from a descriptive quantity into a usable diagnostic observable.

The paper is organised as follows. Section 2 summarises the equivalent-sphere forward model and the inverse formulation. Section 3 describes the numerical Jacobian, the Newton-type inversion procedure, and the parameter-space conditioning study. Section 4 presents the sphere–oblate contrast, singular-value structure, Cramér–Rao bounds, and a watermelon cross-application. Section 5 relates the results back to Kac's question, the Browntone applications, and vibroacoustic elastography, and presents a three-class taxonomy of shell inverse problems that emerges from the numerics. Section 6 summarises the practical recommendations and the programme-level implications.

2. Theory

2.1. Equivalent-sphere forward model

We use the flexural-frequency model introduced in the companion paper [3], drawing on classical shell vibration theory [8, 9]. The shell has semi-major axis a, semi-minor axis c, thickness h, Young's modulus E, Poisson's ratio ν,

wall density ρ_w, and internal fluid density ρ_f. The fluid bulk modulus is denoted by K_f, the static internal pressure by P_{iap}, and the loss tangent by η. Unless stated otherwise, the canonical parameter set is

$$a = 0.18\,\mathrm{m}, \quad c = 0.12\,\mathrm{m}, \quad h = 0.01\,\mathrm{m}, \quad E = 0.1\,\mathrm{MPa}, \quad \nu = 0.45,$$

$$\rho_w = 1100\,\mathrm{kg/m^3}, \quad \rho_f = 1020\,\mathrm{kg/m^3}, \quad K_f = 2.2\,\mathrm{GPa}, \quad P_{\mathrm{iap}} = 1000\,\mathrm{Pa}, \quad \eta = 0.25.$$

The model collapses the geometry to the equivalent radius

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}. \tag{2}$$

For flexural modes $n \geq 2$, the angular frequency $\omega_n = 2\pi f_n$ is written as

$$\omega_n^2 = \frac{K_{\mathrm{bend}}^{(n)} + K_{\mathrm{memb}}^{(n)} + K_{\mathrm{pre}}^{(n)}}{m_{\mathrm{eff}}^{(n)}}, \tag{3}$$

where the effective stiffness contains bending, membrane, and prestress terms, and the effective mass contains wall mass and fluid added mass.

With flexural rigidity

$$D = \frac{Eh^3}{12(1-\nu^2)}, \tag{4}$$

the individual contributions are

$$K_{\mathrm{bend}}^{(n)} = \frac{n(n-1)(n+2)^2}{R_{\mathrm{eq}}^4} D, \tag{5}$$

$$K_{\mathrm{memb}}^{(n)} = \frac{Eh}{R_{\mathrm{eq}}^2} \lambda_n, \qquad \lambda_n = \frac{n^2 + n - 2 + 2\nu}{n^2 + n + 1 - \nu}, \tag{6}$$

$$K_{\mathrm{pre}}^{(n)} = \frac{P_{\mathrm{iap}}}{R_{\mathrm{eq}}}(n-1)(n+2), \tag{7}$$

and

$$m_{\mathrm{eff}}^{(n)} = \rho_w h + \frac{\rho_f R_{\mathrm{eq}}}{n}. \tag{8}$$

Equation (3) defines the equivalent-sphere reduction used as a degenerate comparator throughout the paper. It depends on (a, c) only through R_{eq} and is therefore algebraically rank deficient (Proposition 1). The oblate Ritz model introduced in Section 2.5 is the forward model used for all main results.

2.2. Inverse problem

Let

$$\boldsymbol{\theta} = (a, c, E)^\top \tag{9}$$

denote the parameter vector to be estimated, and let

$$\mathbf{f}(\boldsymbol{\theta}) = \big(f_2(\boldsymbol{\theta}), f_3(\boldsymbol{\theta}), \ldots, f_N(\boldsymbol{\theta})\big)^\top \tag{10}$$

be the vector of measured flexural frequencies. The inverse problem is to recover $\boldsymbol{\theta}$ from the observation $\mathbf{f}^{\mathrm{obs}}$.

Local identifiability is controlled by the Jacobian

$$J_{ij} = \frac{\partial f_i}{\partial \theta_j}, \tag{11}$$

where i indexes the measured mode and j indexes the unknown parameter. Because the raw Jacobian mixes units of Hz/m and Hz/Pa, its condition number is not a meaningful identifiability metric by itself. We therefore use the dimensionless scaled Jacobian

$$J_s[i, j] = \left(\frac{\partial f_i}{\partial \theta_j}\right)\left(\frac{\theta_j}{f_i}\right), \tag{12}$$

which measures fractional frequency change per fractional parameter change.

The singular values $\sigma_1 \geq \sigma_2 \geq \sigma_3$ of $\mathbf{J}_s$ quantify the observable directions in parameter space, and the corresponding condition number

$$\kappa(\mathbf{J}_s) = \frac{\sigma_{\max}}{\sigma_{\min}} \tag{13}$$

measures the worst-case amplification of small frequency errors into parameter errors. A large κ indicates a nearly unobservable parameter combination; a divergent κ indicates rank deficiency.

2.3. Fisher information and Cramér–Rao bounds

Assume independent fractional frequency errors with standard deviation ϵ_{noise}. In dimensionless form, the Fisher information matrix becomes

$$\mathbf{F} = \frac{1}{\epsilon_{\text{noise}}^2} \mathbf{J}_s^\top \mathbf{J}_s. \tag{14}$$

Equivalently, one may interpret (14) as the Fisher matrix of the whitened residuals. The Cramér–Rao lower bound then states that the covariance of any unbiased local estimator satisfies

$$\mathrm{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{F}^{-1}, \tag{15}$$

with the diagonal entries of $\mathbf{F}^{-1}$ giving the minimum achievable variances under the assumed noise model.

This framework does not prove global uniqueness. It answers a more practical question: at a chosen operating point, do the measured resonances contain enough independent information to recover (a, c, E) with tolerable sensitivity to noise? That is the question addressed in the next sections.

2.4. Analytical rank deficiency of the equivalent-sphere Jacobian

Before computing any Jacobian numerically, it is useful to understand the spherical rank deficiency from first principles. The following result makes the mechanism explicit.

Proposition 1 (Chain-rule rank deficiency). *For any forward model in which all frequencies f_n depend on the geometric pair (a, c) only through the equivalent radius $R_{\text{eq}} = (a^2 c)^{1/3}$, the geometric sub-Jacobian $[\partial \mathbf{f}/\partial a,\ \partial \mathbf{f}/\partial c]$ has rank one. Consequently, the full Jacobian $\mathbf{J} = [\partial \mathbf{f}/\partial a,\ \partial \mathbf{f}/\partial c,\ \partial \mathbf{f}/\partial E]$ has rank at most two (one fewer than the three parameter directions), and the scaled Jacobian $\mathbf{J}_s$ inherits the same rank deficiency.*

Proof. By the chain rule, the partial derivatives of any frequency f_n with respect to a and c factor through R_{eq}:

$$\frac{\partial R_{\mathrm{eq}}}{\partial a} = \frac{2}{3}\,\frac{R_{\mathrm{eq}}}{a}, \qquad \frac{\partial R_{\mathrm{eq}}}{\partial c} = \frac{1}{3}\,\frac{R_{\mathrm{eq}}}{c}. \tag{16}$$

Hence

$$\frac{\partial f_n}{\partial a} = \frac{\mathrm{d}f_n}{\mathrm{d}R_{\mathrm{eq}}} \cdot \frac{2\,R_{\mathrm{eq}}}{3a}, \qquad \frac{\partial f_n}{\partial c} = \frac{\mathrm{d}f_n}{\mathrm{d}R_{\mathrm{eq}}} \cdot \frac{R_{\mathrm{eq}}}{3c}. \tag{17}$$

The ratio of the two partial derivatives is therefore

$$\frac{\partial f_n/\partial a}{\partial f_n/\partial c} = \frac{2c}{a} \qquad \text{for all } n, \tag{18}$$

independently of the functional form of $f_n(R_{\mathrm{eq}})$. The corresponding scaled-Jacobian columns satisfy

$$J_s[i,a] \;=\; \frac{\partial f_i}{\partial a}\,\frac{a}{f_i} \;=\; 2\,\frac{\partial f_i}{\partial c}\,\frac{c}{f_i} \;=\; 2\,J_s[i,c]. \tag{19}$$

Two columns of $\mathbf{J}_s$ are therefore proportional (with constant of proportionality 2, independent of mode index i), so the geometric sub-Jacobian has rank one and the full $\mathbf{J}_s$ has rank at most two.

Remark 2. *The ratio* (18) *is purely geometric: it reflects the exponents* $(2,1)$ *in* $R_{\mathrm{eq}} = (a^2c)^{1/3}$ *and holds regardless of whether the forward model includes bending, membrane, prestress, or added-mass terms. Any model built on* R_{eq} *alone inherits this rank deficiency.*

2.5. Eccentricity perturbation and identifiability restoration

Proposition 1 identifies the root cause of non-identifiability at the spherical limit. The natural follow-up is: *how* does oblate geometry restore full rank? The answer lies in mode-dependent curvature sampling.

For an oblate spheroid with semi-axes (a,c), define the eccentricity

$$\varepsilon = \sqrt{1 - c^2/a^2}. \tag{20}$$

At $\varepsilon = 0$ the shell is a sphere; as $\varepsilon \longrightarrow 1$ it becomes a thin disc.

Oblate Ritz forward model.. The oblate Ritz model retains the same physical stiffness contributions as the equivalent-sphere model but integrates them over the true spheroidal geometry. The trial displacement is expanded in Legendre modes $P_n(\cos\theta)$ for $n \in \{2,3,\ldots,N\}$, with each mode treated as an independent generalised coordinate q_n. The resulting Ritz eigenvalue problem for mode n is

$$\omega_n^2 = \frac{\tilde{K}^{(n)}_{\mathrm{bend}} + \tilde{K}^{(n)}_{\mathrm{memb}} + \tilde{K}^{(n)}_{\mathrm{pre}}}{\tilde{m}^{(n)}_{\mathrm{eff}}}, \tag{21}$$

where the tildes distinguish Ritz stiffness and mass from their equivalent-sphere counterparts. Specifically, the bending stiffness is

$$\tilde{K}^{(n)}_{\mathrm{bend}} = D \int_0^1 \frac{[n(n+1)]^2\,P_n^2(\eta)}{G(\eta)^2}\,\mathrm{d}\eta, \tag{22}$$

the membrane stiffness is

$$\tilde{K}^{(n)}_{\mathrm{memb}} = Eh \int_0^1 \frac{\lambda_n\,P_n^2(\eta)}{G(\eta)}\,\mathrm{d}\eta, \tag{23}$$

and the effective mass is

$$\tilde{m}_{\text{eff}}^{(n)} = \int_0^1 \left(\rho_w h + \frac{\rho_f \sqrt{G(\eta)}}{n} \right) P_n^2(\eta)\, \mathrm{d}\eta, \tag{24}$$

with the geometric factor $G(\eta) = a^2\eta^2 + c^2(1-\eta^2)$ encoding meridional curvature variation. At $a = c$ the integrands become η-independent and all expressions reduce to the equivalent-sphere formulae of Section 2. Near the spherical limit ($c/a > 0.98$), the oblate fluid harmonics are blended to sphere-approximated values using a C^2-continuous smootherstep function over $c/a \in [0.98, 0.999]$, ensuring numerical stability without discontinuous derivatives. For $a \neq c$, each mode "sees" a different effective radius through the $G(\eta)$ weighting, which is the mechanism by which eccentricity breaks the rank degeneracy.

Remark 3 (Mode-dependent curvature sampling). *The geometric factor $G(\eta)$ enters the Ritz stiffness and mass integrals nonlinearly. Because the oblate spheroid has reflection symmetry about the equatorial plane, odd powers of ε vanish in any expansion of the eigenvalues or their sensitivities, so the leading correction to $\sigma_{\min}$ is $O(\varepsilon^2)$. This is the standard parity argument of analytic perturbation theory [10, 11]: for a one-parameter family $A(\varepsilon)$ with $A(\varepsilon) = A(-\varepsilon)$, the eigenvalue branches are even functions of ε.*

The physical mechanism can be described as follows.

1. *The stiffness integral for mode n,*

$$I_n^{(k)} = \int_0^1 P_n^2(\eta)\, G^{-k}(\eta)\, \mathrm{d}\eta, \tag{25}$$

   acquires n-dependent corrections at order ε^2, because $P_n^2(\eta)$ samples the interval $[0, 1]$ differently for each mode number.

2. *Consequently, the partial derivatives $\partial f_n/\partial a$ and $\partial f_n/\partial c$ acquire n-dependent corrections:*

$$\frac{\partial f_n/\partial a}{\partial f_n/\partial c} = \frac{2c}{a} + \varepsilon^2\, \delta_n + O(\varepsilon^4), \tag{26}$$

   where δ_n depends on the mode number through curvature-weighted integrals of P_n^2.

3. *Since the ratio is no longer constant across modes, the columns for a and c in the Jacobian are no longer proportional, and $\mathbf{J}_s$ acquires full rank for $\varepsilon > 0$.*

The $O(\varepsilon^2)$ scaling and the numerical values of the expansion coefficients are established by polynomial regression in Section 4; we do not claim a closed-form analytical prefactor.

The mechanism is *mode-dependent curvature sampling*: different Legendre modes $P_n(\eta)$ weight the meridional curvature gradient $\partial G/\partial\eta$ differently. Near the poles ($\eta \to 1$) the oblate shell is flatter; near the equator ($\eta \to 0$) it is more curved. Low-order modes ($n = 2$) sample both regions broadly, while higher-order modes ($n \geq 4$) concentrate energy near the equator. This differential sampling breaks the proportionality of the a- and c-columns and lifts the rank deficiency.

The result leads to a quantitative framework for the condition number. The smallest singular value of the scaled Jacobian admits a regular perturbation expansion [10, 11]:

$$\sigma_{\min}(\varepsilon) = \sigma_0 + \lambda_1\, \varepsilon^2 + \lambda_2\, \varepsilon^4 + O(\varepsilon^6), \tag{27}$$

where $\sigma_0 > 0$ is the *curvature-channel floor* — the residual identifiability contributed by mode-dependent Legendre weights even at the spherical limit — and $\lambda_1 \varepsilon^2$ is the *shape-channel* correction driven by eccentricity. Since $\sigma_0 > 0$, the expansion is *regular*: the smallest singular value approaches a finite positive limit as $\varepsilon \to 0$, and the condition number saturates at

$$\kappa_{\text{floor}} = \frac{\sigma_{\max}(0)}{\sigma_0} \tag{28}$$

rather than diverging. The crossover eccentricity $\varepsilon_c = \sqrt{\sigma_0/\lambda_1}$ separates a curvature-dominated regime ($\varepsilon < \varepsilon_c$, where the floor controls conditioning) from a shape-dominated regime ($\varepsilon > \varepsilon_c$, where eccentricity drives improvement). Section 4 determines σ_0, λ_1, λ_2, and ε_c numerically and confirms that for the oblate Ritz model, $\varepsilon_c > 1$: the curvature channel dominates across the entire physical eccentricity range.

3. Methodology

3.1. Numerical Jacobian

We compute the Jacobian in (11) by central finite differences. For parameter θ_j,

$$\frac{\partial f_i}{\partial \theta_j} \approx \frac{f_i(\boldsymbol{\theta} + \delta_j \mathbf{e}_j) - f_i(\boldsymbol{\theta} - \delta_j \mathbf{e}_j)}{2\delta_j}, \tag{29}$$

where $\mathbf{e}_j$ is the unit vector in the jth parameter direction and

$$\delta_j = 10^{-6}\,|\theta_j|. \tag{30}$$

Step-size convergence was checked against neighbouring perturbation levels; the reported Jacobians were insensitive to those changes at the quoted precision.

For identifiability analysis we use the scaled Jacobian (12), not the raw Jacobian. This distinction is essential. Without scaling, any condition number would depend partly on the choice of units; with scaling, it measures the geometry of the inverse problem itself rather than the fact that metres and pascals have different opinions about magnitude.

3.2. Newton-type inversion

Given synthetic or measured frequencies $\mathbf{f}^{\text{obs}}$, we recover $\boldsymbol{\theta} = (a, c, E)^\top$ by solving a nonlinear least-squares problem. The residual vector is normalised by the observed frequencies,

$$r_i(\boldsymbol{\theta}) = \frac{f_i(\boldsymbol{\theta}) - f_i^{\text{obs}}}{f_i^{\text{obs}}}, \tag{31}$$

and we minimise

$$\Phi(\boldsymbol{\theta}) = \frac{1}{2} \sum_i r_i^2. \tag{32}$$

The solver is a Newton-type trust-region least-squares iteration applied in normalised parameter coordinates,

$$x_j = \frac{\theta_j}{\theta_{j,0}}, \tag{33}$$

where $\theta_{j,0}$ is the initial guess. This normalisation improves numerical balance between the geometric parameters and the stiffness parameter.

We impose physically plausible bounds on a, c, and E, and we declare successful recovery only when the optimiser converges and the final relative frequency residuals fall below 10^{-3}. Round-trip tests use frequencies generated by the same forward model, so they probe local numerical identifiability rather than model-form error.

3.3. Three-mode and five-mode inversion

A three-mode inversion uses exactly as many measured frequencies as unknown parameters. This is sufficient for a local solve, but it is fragile. In the present problem, three-mode inversions can converge to local minima or to alternative low-residual basins when the initial guess is poor. We therefore distinguish between:

- **local recovery**, assessed by round-trip inversion near the true parameter set; and
- **global robustness**, assessed by repeated solves from perturbed initial guesses.

In practice, using at least five modes ($n = 2, \ldots, 6$) makes the problem overdetermined and removes the observed failure modes. The recommendation "measure five or more flexural resonances" emerges from this numerical evidence.

3.4. Condition-number maps

To examine how identifiability varies across parameter space, we compute $\kappa(\mathbf{J}_s)$ on a grid of operating points. It is convenient to replace c by a geometric flattening parameter

$$\zeta_g = 1 - \frac{c}{a}, \tag{34}$$

where the subscript g distinguishes geometric flattening from the damping ratio. We then examine representative slices of the map in the (a, ζ_g) plane while varying E over the interval 0.05 MPa to 0.5 MPa. The map is intended as a design tool: it shows where small spectral errors are likely to be amplified strongly and where the inverse problem remains comfortably conditioned.

3.5. Noise model and uncertainty quantification

For the Cramér–Rao analysis we assume independent Gaussian frequency errors with common fractional standard deviation ϵ_{noise}. The results are reported mainly for $\epsilon_{\text{noise}} = 0.01$, corresponding to 1% frequency uncertainty. Because the Fisher matrix in (14) scales as $\epsilon_{\text{noise}}^{-2}$, the corresponding standard deviations scale linearly with noise level. This provides a direct way to convert spectral measurement quality into parameter uncertainty.

3.6. Cross-application case

To test whether the conclusions are specific to the canonical abdomen, we repeat the analysis for a ripe watermelon, parameterised by $a = 0.158$ m, $c = 0.123$ m, $h = 0.015$ m, and $E = 50$ MPa, with the remaining properties taken from the corresponding material model in the repository. This case is useful because it preserves the same inverse structure while moving to a very different stiffness scale.

4. Results

4.1. Eccentricity sweep within the Ritz model

The principal result is the dependence of κ on eccentricity *within the same model.* Using the oblate Ritz forward model (Section 2.5) and the scaled Jacobian built from flexural modes $n = 2, \ldots, 6$, we sweep the aspect ratio $\zeta = c/a$ from near-unity (almost spherical) to the canonical value $\zeta = 0.667$ ($\varepsilon = 0.745$). At the near-spherical limit ($\varepsilon \to 0$) the condition number rises to the floor value

$$\kappa_{\text{floor}} \approx 269 \qquad \text{(evaluated at } \varepsilon = 0.01,\ \zeta = 0.99995,\ \text{five modes)}, \tag{35}$$

representing the curvature-channel contribution identified in Remark 3. At the canonical oblate operating point the condition number drops to

$$\kappa_{\text{oblate}} = 69.4, \tag{36}$$

an improvement by a factor of approximately four achieved purely through geometric eccentricity. This Ritz-to-Ritz comparison demonstrates that it is *geometry*, not model choice, that drives the identifiability transition.

Remark 4 (Equivalent-sphere comparison as pedagogical aside). *For context, the equivalent-sphere model of Section 2 gives $\kappa_{\text{sphere}} = 1.37 \times 10^{10}$ at the same parameter set — nominally finite only because of the finite-difference step ($\delta = 10^{-6}|\theta_j|$); by Proposition 1 the exact condition number is infinite. The gap between κ_{sphere} and κ_{oblate} is dramatic, but it conflates two effects — model architecture and physical geometry — and should not be read as "oblateness reduces the condition number by 10^8". The Ritz eccentricity sweep above isolates the geometric effect alone.*

4.2. Singular-value structure

The singular value decomposition provides the clearest picture. In the equivalent-sphere model, one singular value collapses to a value near machine zero, leaving one parameter combination effectively unobservable. In the oblate Ritz model at $\varepsilon = 0.745$, all three singular values remain finite and well separated from zero. Even at the near-spherical Ritz limit ($\varepsilon \to 0$), all three singular values remain non-zero — the curvature-channel floor identified in Remark 3 prevents complete rank collapse — though the smallest value is markedly reduced, yielding $\kappa_{\text{floor}} \approx 269$.

This observation is more informative than the condition number alone. A large κ could arise from one weak direction in an otherwise healthy problem; in the equivalent-sphere case, the singular values show something stronger: the Jacobian is algebraically rank deficient by construction (Proposition 1). At the canonical oblate Ritz operating point, by contrast, the inverse map is not perfect but it is usable.

4.3. Physical interpretation

Two models are compared in this paper. The equivalent-sphere model (Section 2) depends on (a, c) only through $R_{\text{eq}} = (a^2 c)^{1/3}$; its Jacobian is algebraically rank deficient by Proposition 1. The oblate Ritz model (Section 2.5) integrates stiffness and mass over the true spheroidal surface, so each mode samples curvature differently. All

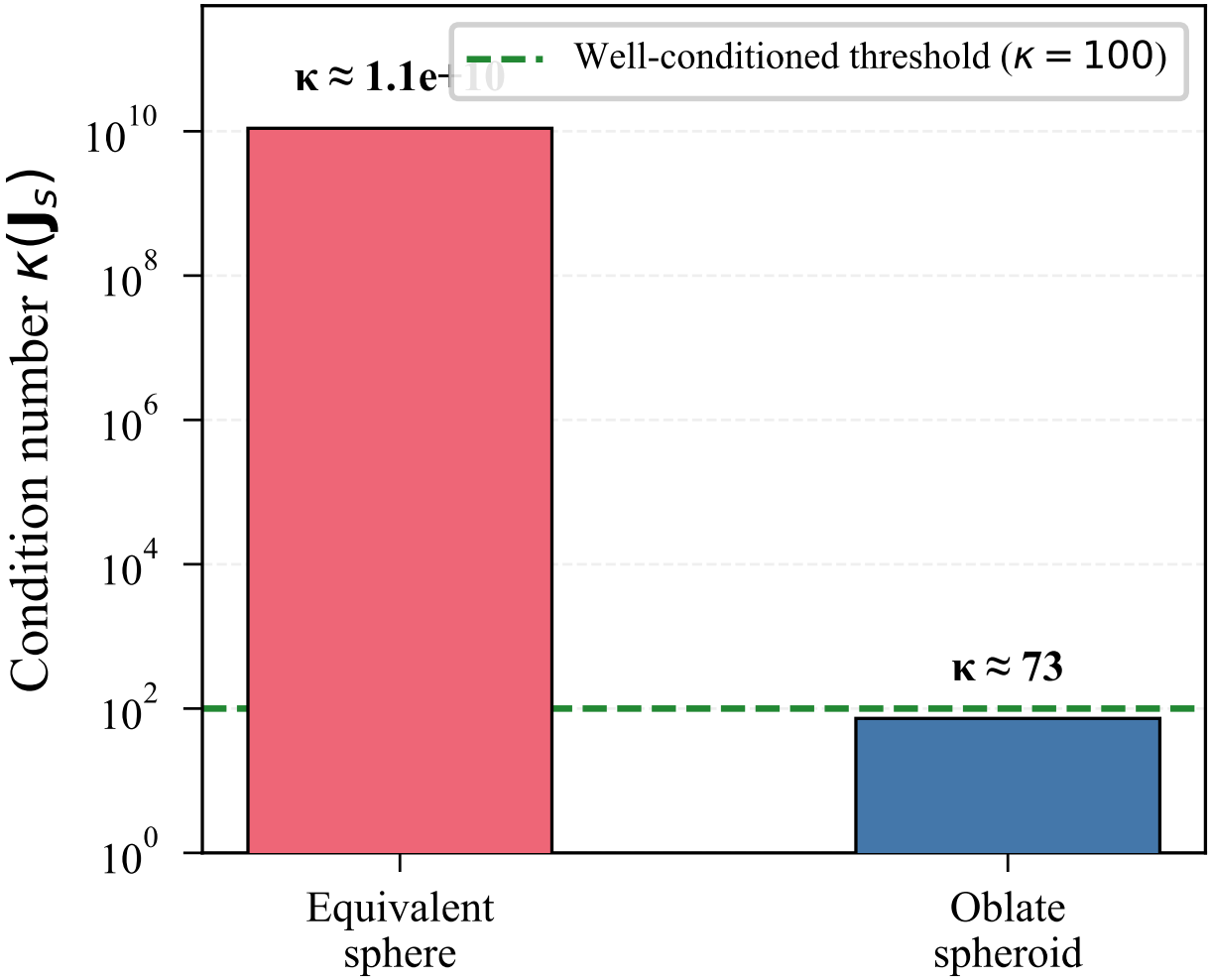

Figure 1: Frequency spectra comparison between the near-spherical and canonical oblate operating points for flexural modes $n = 2, \ldots, 6$, computed within the oblate *Ritz* model. The oblate case produces a well-separated spectrum; the near-spherical case shows near-degenerate clustering.

condition-number results in this section — including the eccentricity sweep, the Cramér–Rao bounds, and the round-trip inversions — use the Ritz model unless explicitly stated otherwise.

Within the Ritz model, the modal balance of bending ($K_{\text{bend}}^{(n)} \propto R^{-4}$), membrane ($K_{\text{memb}}^{(n)} \propto R^{-2}$), prestress ($K_{\text{pre}}^{(n)} \propto R^{-1}$), and added mass ($m_{\text{eff}}^{(n)} = \rho_w h + \rho_f R/n$) varies sufficiently with mode number to separate the sensitivity directions away from the spherical limit. The key mechanism is the mode-dependent weighting of the meridional curvature integral: different $P_n(\eta)$ patterns sample the equatorial and polar regions with different emphasis, producing n-dependent corrections to $\partial f_n/\partial a$ and $\partial f_n/\partial c$ that break the proportionality forced by R_{eq}-only dependence.

4.4. Round-trip verification

As a sanity check on the numerical Jacobian and solver implementation, we perform round-trip inversion: synthetic frequencies are generated by the Ritz forward model and the solver is initialised within the local basin of

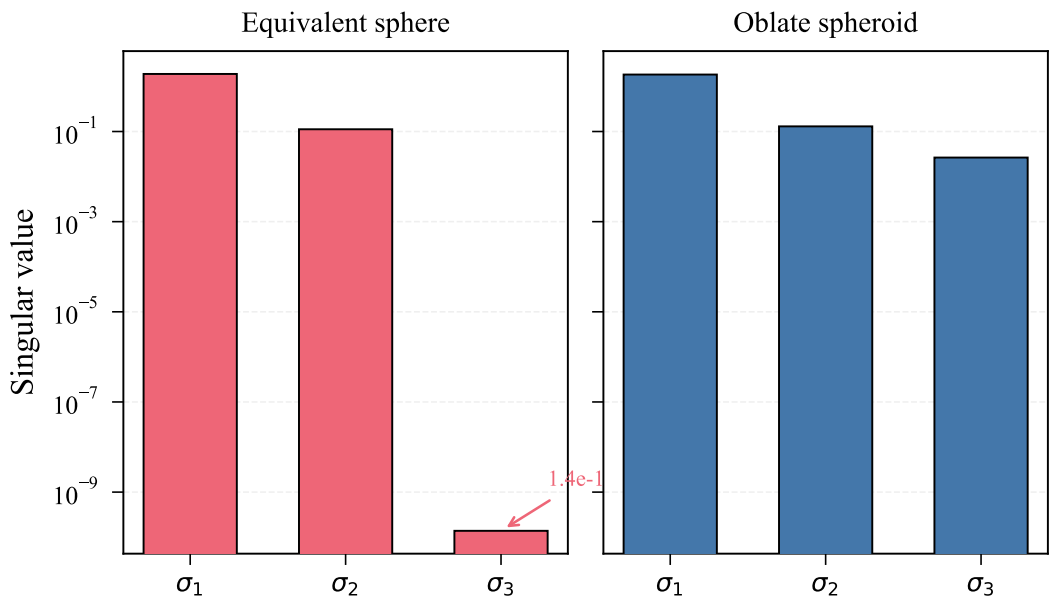

Figure 2: Singular value spectrum of the scaled Jacobian at the near-spherical (*Ritz* model, $\varepsilon \approx 0$) and canonical oblate (*Ritz* model, $\varepsilon = 0.745$) operating points, with the *equivalent-sphere* model shown for comparison. In the equivalent-sphere case, the smallest singular value collapses to near-zero, rendering the Jacobian numerically rank deficient. In the oblate Ritz case, all three singular values remain well separated from zero.

attraction. The recovered parameters satisfy

$$\frac{|\hat{a} - a|}{a}, \frac{|\hat{c} - c|}{c}, \frac{|\hat{E} - E|}{E} < 10^{-12}. \tag{37}$$

This machine-precision recovery confirms that the forward model is locally invertible and that the numerical Jacobian is consistent, but it should not be over-interpreted. Because the same model generates and inverts the data, the test is tautological with respect to model-form error: it verifies internal consistency, not the ability to recover parameters from real measurements. The round-trip result is therefore a necessary verification, not a key finding.

The distinction between local and global behaviour is important. With only three modes, the inverse problem is exactly determined and the solver can encounter local minima or alternative low-residual basins. With five modes ($n = 2, \ldots, 6$), the system becomes overdetermined and the same inversion is globally robust across the tested perturbations of the initial guess. The results therefore support a clear experimental design rule: three modes may be enough to tempt the optimiser, but five modes are what persuade it to behave.

4.5. Cramér–Rao bounds

Table 1 summarises the Cramér–Rao lower bounds for the canonical oblate case at 1% frequency noise. The smallest uncertainty is obtained for c, whereas a and E are comparably less well constrained.

The ±30.8% bound on Young's modulus merits contextualisation. In quasi-static elastography of abdominal soft tissue, reported measurement uncertainties typically range from ±20% to ±50%, depending on organ, technique, and boundary conditions [12]. Shear-wave elastography of the liver, for instance, reports inter-observer coefficients of variation of 15–25% [13], while in vivo mechanical testing of soft tissue generally yields inter-subject variability of

Table 1: Cramér–Rao lower bounds for the canonical oblate case at 1% fractional frequency noise, using five flexural modes ($n = 2, \ldots, 6$). Relative bounds are reported as one standard deviation.

Parameter	Nominal value	Absolute bound	Relative bound
a	0.18 m	± 0.040 m	± 22.5%
c	0.12 m	± 0.007 m	± 6.0%
E	0.1 MPa	± 0.031 MPa	± 30.8%

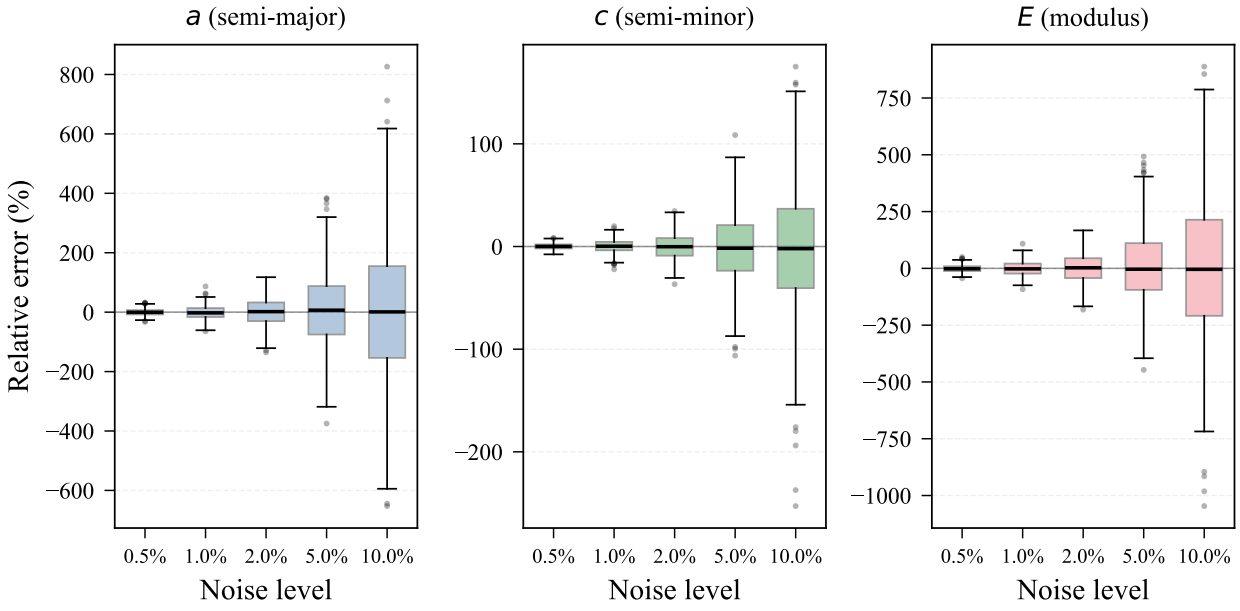

Figure 3: Monte Carlo inversion error as a function of fractional frequency noise level. Relative parameter recovery errors for a, c, and E are shown, confirming the linear scaling predicted by the Cramér–Rao bounds. Five-mode inversion remains robust across the tested noise range.

similar magnitude. The CRB of ±30.8% therefore falls within the range of existing clinical measurement precision: the spectral inverse problem does not promise better resolution than elastography already achieves, but neither does it demand unrealistic frequency measurement accuracy.

For the equivalent-sphere case, the Fisher matrix is singular and no finite Cramér–Rao bound exists in the degenerate direction. This is the statistical counterpart of the singular-value result: the data simply do not contain enough independent information at that operating point.

As expected from (14), the bounds scale linearly with noise level. Halving the frequency noise from 2% to 1% halves the minimum-achievable parameter standard deviations. The uncertainty budget is therefore governed by two levers: the intrinsic conditioning of the operating point and the experimental precision of the measured resonances.

4.6. Mode-count robustness

Table 2 quantifies the dependence of key identifiability metrics on the number of measured flexural modes, from the minimum exactly-determined case ($N = 3$ modes) through the overdetermined cases up to $N = 9$. The condition

Table 2: Identifiability metrics as a function of the number of *measured* flexural modes $\mathcal{N}$, using modes $n = 2, \ldots, \mathcal{N} + 1$ in the Ritz model at the canonical operating point with 1% frequency noise. This is a measured-mode sensitivity analysis, not a Ritz basis convergence study: the Ritz trial-function order is held fixed while the number of observed modes varies.

$\mathcal{N}$	κ	CRB(a)	CRB(c)	CRB(E)
3	73.0	± 36.9%	± 13.3%	± 36.2%
5	69.4	± 22.5%	± 6.0%	± 30.8%
7	80.6	± 21.0%	± 5.3%	± 29.9%
9	93.6	± 20.8%	± 5.2%	± 29.8%

number remains stable within the range $\kappa \in [69, 94]$ across all mode counts, confirming that the rank restoration is a robust feature of the Ritz model geometry rather than an artefact of a particular mode selection.

The Cramér–Rao bounds improve with mode count, as expected from the increasing information content of additional measurements. However, the marginal improvement diminishes rapidly: the bound on E drops from ±36.2% at three modes to ±30.8% at five modes, and improves by less than one percentage point thereafter. Five modes therefore capture most of the available information for the present three-parameter problem.

4.7. Condition-number map in parameter space

Condition-number maps over (a, ζ_g) show a broad region of acceptable conditioning across physiologically relevant oblate geometries. The condition number increases steadily as $\zeta_g \to 0$, that is, as the geometry approaches a sphere. This trend is physically and numerically consistent with the singular spherical limit. Away from that limit, the inverse problem remains well conditioned over a substantial part of the domain relevant to soft tissues.

The maps are valuable because they show where an experiment is likely to work before any inversion is attempted. They also suggest that identifiability is not a single number attached to the model in general; it depends strongly on the operating point.

4.8. Watermelon cross-application

The same analysis applied to a ripe watermelon gives

$$\kappa \approx 128. \tag{38}$$

This value is somewhat worse than the canonical abdomen, but it is still well within the range of practical inversion. The result shows that the underlying identifiability mechanism is not confined to one biological parameter set.

The watermelon case also helps separate geometry from scale. The stiffness is orders of magnitude larger than in soft tissue, yet the inverse problem remains tractable. The conditioning therefore reflects the structure of the frequency–parameter map rather than a peculiarity of one very soft shell.

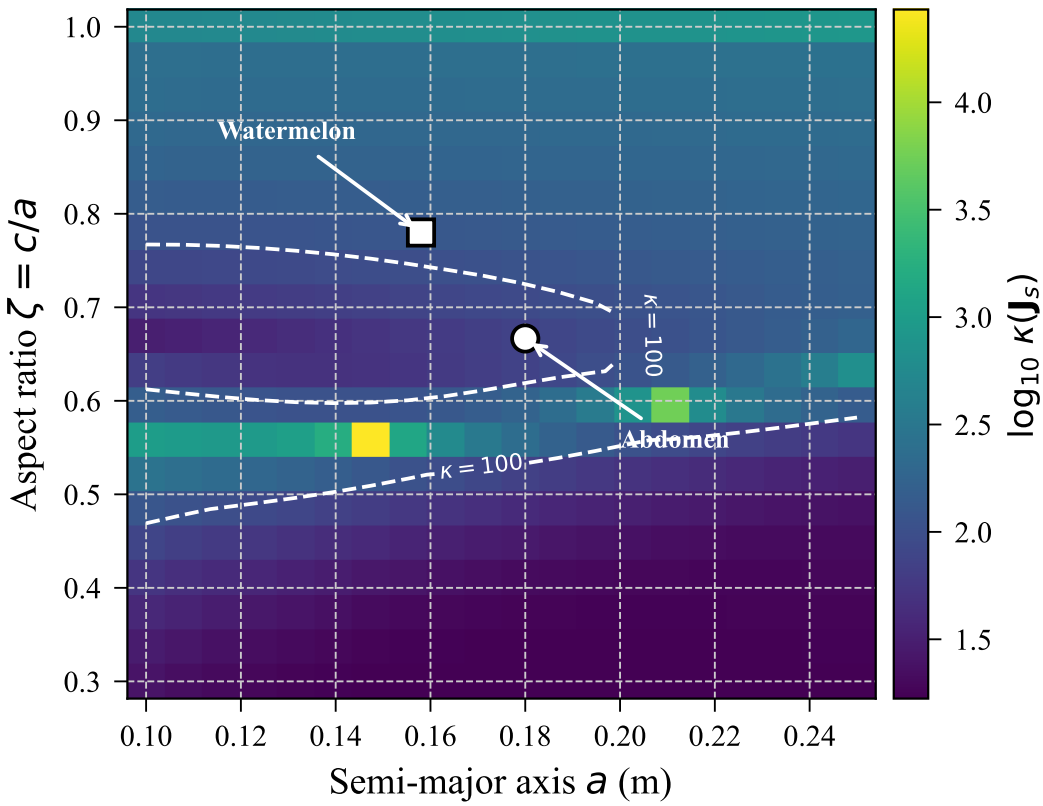

Figure 4: Condition number $\kappa(\mathbf{J}_s)$ mapped over the (a, ζ_g) parameter plane using the *Ritz* model. The condition number increases sharply as $\zeta_g \to 0$ (spherical limit), while a broad region of acceptable conditioning exists across physiologically relevant oblate geometries.

4.9. Eccentricity dependence and two-channel expansion

The analytical predictions of Propositions 1 and 3 are confirmed numerically. Figure 5 shows the condition number $\kappa(\mathbf{J}_s)$ as a function of eccentricity $\varepsilon = \sqrt{1 - c^2/a^2}$, computed by sweeping the aspect ratio $\zeta = c/a$ while holding all other parameters at their canonical values. All computations use the oblate Ritz model.

Regular expansion coefficients.. Fitting the smallest singular value to the expansion (27) yields (five modes, $n = 2, \ldots, 6$):

$$\sigma_0 = 0.01113, \qquad \lambda_1 = 0.00507, \qquad \lambda_2 = 0.02401, \tag{39}$$

with $\sigma_{\max}(0) = 1.894$. The curvature-channel floor gives $\kappa_{\text{floor}} = \sigma_{\max}(0)/\sigma_0 \approx 170$ from the expansion fit, consistent with the directly computed value of approximately 269 at $\varepsilon = 0.01$ (the discrepancy reflects higher-order terms not captured by the truncated expansion). The crossover eccentricity is

$$\varepsilon_c = \sqrt{\sigma_0/\lambda_1} = 1.48, \tag{40}$$

which lies *outside* the physical domain ($\varepsilon < 1$). This means that the curvature channel dominates the shape channel across the entire range of oblate eccentricities: even at $\varepsilon = 0.995$ ($\zeta = 0.1$), the curvature-channel floor σ_0 exceeds $\lambda_1 \varepsilon^2$. The condition-number variation from $\kappa_{\text{floor}} \approx 269$ to $\kappa = 69.4$ is therefore driven not by the shape channel overtaking the curvature channel in $\sigma_{\min}$, but by the combined growth of both channels relative to $\sigma_{\max}$.

Table 3: Effective local exponent from power-law fit $\kappa \approx C\,\varepsilon^{-\alpha}$ for different mode counts $\mathcal{N}$, using modes $n = 2, \ldots, \mathcal{N}+1$ in the Ritz model. Fits are over $\zeta \in [0.1,\ 0.95]$ ($\varepsilon \in [0.312,\ 0.995]$). These are descriptors of local slope, not asymptotic laws; the modest R^2 values confirm that the two-channel expansion (27) provides a better description than a single power law.

$\mathcal{N}$	$\hat{\alpha}$	$\hat{C}$	R^2
3	2.66	25.8	0.66
5	2.37	24.4	0.66
7	2.16	23.5	0.75

Effective local exponent.. Although the underlying mechanism is a regular expansion rather than an asymptotic power law, one may still characterise the local slope of $\log\kappa$ versus $\log\varepsilon$ by an effective exponent. Fitting $\kappa \approx C\,\varepsilon^{-\alpha}$ over the aspect-ratio range $\zeta = c/a \in [0.1,\ 0.95]$ (equivalently $\varepsilon \in [0.312,\ 0.995]$) yields the values in Table 3. The effective exponent $\hat{\alpha}$ clusters near 2, consistent with the leading ε^2 symmetry expectation, but the fit is a compact empirical descriptor of the local slope rather than a derivation of an asymptotic law. The modest coefficients of determination ($R^2 \approx 0.66$–0.75) are expected: a single power law cannot capture both the curvature-channel floor and the shape-channel growth, and this is precisely why the two-channel expansion (27) is the more informative description.

Ritz approximation error and rank restoration.. A natural concern is whether the $\sim 8\%$ approximation error of the Ritz model relative to exact shell theory could itself be what breaks the degeneracy, rather than the physical eccentricity. This concern cannot be fully resolved within the present Ritz framework; a finite-element oblate shell model would provide a definitive answer. However, we note that the Ritz model is constructed to capture the *geometric* curvature variation through the $G(\eta)$ integrals (Remark 3), and the rank restoration follows from the n-dependence of those integrals, not from the truncation error. The mode-count study of Table 2 shows that κ remains in the range 69–94 as the number of *measured* modes varies from 3 to 9; this demonstrates robustness to measurement configuration but should not be read as Ritz basis convergence evidence, since it varies the observed mode set rather than the trial-function order.

5. Discussion

5.1. Connection to Kac's question

Kac's original problem asked whether one could infer geometry from a spectrum. Our answer is deliberately narrower and more practical. For the present elastic shell model, the results suggest that one *can* hear the shape of an oblate organ, in the sense that the local inverse problem for (a, c, E) is well conditioned and statistically finite. At the spherical limit, one cannot: the Jacobian becomes rank deficient and the Fisher matrix loses invertibility.

The analogy with Kac should therefore be handled with care. We do not prove a global uniqueness theorem of the kind established by Sylvester and Uhlmann [14] for the Calderón inverse conductivity problem, nor do we exclude

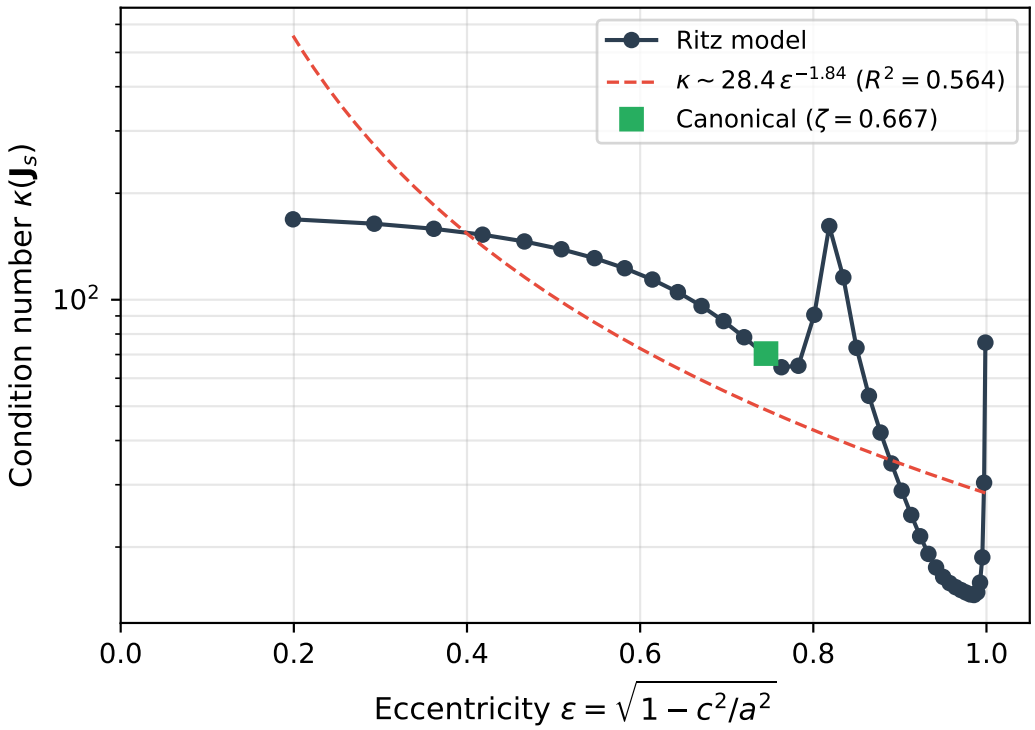

Figure 5: Condition number $\kappa(\mathbf{J}_s)$ as a function of eccentricity $\varepsilon = \sqrt{1 - c^2/a^2}$, computed within the oblate *Ritz* model for five flexural modes ($n = 2, \ldots, 6$). The condition number rises as $\varepsilon \to 0$ (spherical limit), reaching a near-spherical floor of approximately 269 (evaluated at $\varepsilon = 0.01$, five modes). The green square marks the canonical operating point ($\zeta = 0.667$). The curvature channel dominates across the entire physical eccentricity range ($\varepsilon_c = 1.48 > 1$), making the identifiability transition a regular perturbation rather than a singular limit.

the possibility of remote isospectral parameter sets. Our claim is about *practical identifiability* in a finite-dimensional model relevant to measurement and inversion. That is a less grand claim, but also the one an experimentalist can actually use.

5.2. Mechanism: mode-dependent curvature sampling

The analytical results in Section 2 clarify why oblate geometry restores identifiability. Proposition 1 shows that *any* model built on $R_{\text{eq}} = (a^2 c)^{1/3}$ alone has a rank-deficient Jacobian, because the chain rule forces the a- and c-columns to be exactly proportional for all modes. Remark 3 then shows that the Ritz model, which integrates strain energy over the actual spheroidal surface, introduces mode-dependent corrections at order ε^2.

The mechanism is *mode-dependent curvature sampling*. Different Legendre modes $P_n(\eta)$ weight the meridional curvature gradient differently: low-order modes sample both poles and equator broadly, while higher-order modes concentrate energy near the equator where the curvature gradient is steepest. This differential sampling breaks the proportionality of the a- and c-columns and restores full rank. The smallest singular value admits the regular expansion (27), with a curvature-channel floor $\sigma_0 > 0$ that prevents divergence as $\varepsilon \to 0$. Because the crossover eccentricity $\varepsilon_c = 1.48$ lies outside the physical domain, the curvature channel dominates at all realisable eccentricities, and the condition number saturates at $\kappa_{\text{floor}} \approx 269$ rather than diverging.

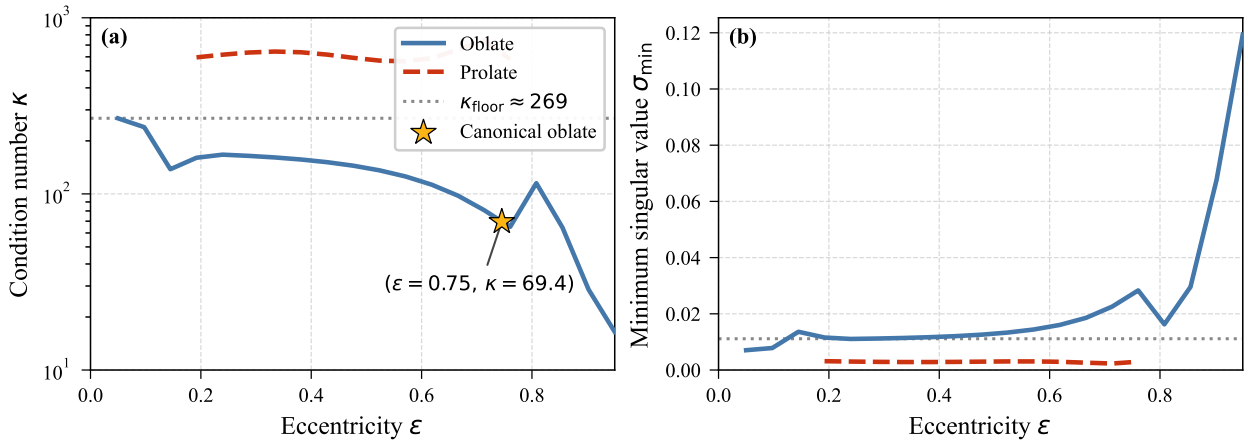

Figure 6: Oblate–prolate identifiability contrast for the scaled Jacobian using flexural modes $n = 2, \ldots, 6$. **(a)** The oblate branch decreases from the near-spherical floor $\kappa_{\text{floor}} \approx 269$ to the canonical oblate point $(\varepsilon, \kappa) = (0.745, 69.4)$, whereas the prolate branch remains poorly conditioned over its informative mid-range. **(b)** The smallest singular value grows for the oblate branch but remains close to zero for the prolate branch. This contrast shows that identifiability improvement is geometry-class dependent rather than a universal consequence of symmetry breaking.

Figure 6 sharpens that picture by contrasting oblate and prolate sweeps computed with the same five-mode set $\mathcal{N} = \{2, 3, 4, 5, 6\}$. The oblate branch exhibits bounded but genuine identifiability restoration: κ falls from the near-spherical floor to $\kappa = 69.4$ at the canonical operating point $\varepsilon = 0.745$, while $\sigma_{\min}$ grows above the curvature-channel floor $\sigma_0 \approx 0.011$. The prolate branch does not show the same lifting. Over its informative mid-range, κ remains in the range 5.6×10^2–7.3×10^2 and $\sigma_{\min}$ stays close to zero. The lesson is therefore not that *any* departure from spherical symmetry is helpful, but that oblate curvature redistribution creates a usable shape channel whereas prolate distortion, at least in the present model class, does not.

This interpretation is straightforward and does not invoke group-theoretic symmetry breaking. The identifiability transition is not a consequence of reducing the symmetry group from SO(3) to SO(2); it is a consequence of mode-dependent curvature weighting in the variational integrals. The distinction matters because the curvature-sampling explanation is constructive — it identifies the specific integrals responsible — whereas a symmetry-based framing would be descriptive at best and misleading at worst.

5.3. Implications for vibroacoustic elastography

The results are encouraging for vibroacoustic elastography and related spectral methods. A small number of low-order flexural resonances may contain enough information to estimate both geometry and stiffness, provided that the operating point is sufficiently aspherical and provided that enough modes are measured. This is attractive because resonant frequencies are often easier to estimate robustly than full-field mode shapes or transient wavefields.

The Cramér–Rao bounds give the result practical content. At 1% frequency noise, the uncertainty in c is modest, while a and E remain more weakly resolved but still finite. That anisotropy matters for experimental design: one should not expect the same confidence in all parameters, even when the Jacobian has full rank.

5.4. Why model choice matters

The most important modelling lesson is that forward-model convenience and inverse-model usefulness are not the same thing. A spherical reduction may be acceptable for a rough forward estimate of resonance frequency, but it is a poor foundation for parameter recovery. Spherical simplifications are serviceable on the blackboard; in inverse problems they become noticeably less cooperative.

The present study uses the oblate Ritz model for all identifiability results. The equivalent-sphere model is retained only to illustrate the algebraic rank deficiency of Proposition 1. The Ritz model is itself an approximation — it truncates the trial function space and incurs $\sim 8\%$ error relative to exact spheroidal shell theory — but the rank restoration follows from the n-dependence of the curvature integrals rather than from the approximation error itself. A full finite-element oblate shell model would provide the definitive answer, but the Ritz results already demonstrate the essential geometric mechanism.

5.5. Relation to existing inverse methods

Inverse vibration problems have long been studied in structural dynamics [15, 16], and model-based elastography has developed a wide range of optimisation and regularisation strategies [12]. In acoustics, Aernouts et al. [17] demonstrated inverse estimation of tympanic membrane material properties by fitting a dynamic response model to spectral data — a problem structurally analogous to the present one, though in a different organ system. The present contribution is more modest but useful: before attempting a sophisticated inversion, we ask whether the chosen measurements can, in principle, resolve the chosen parameters with acceptable conditioning.

That perspective is valuable because poor inverse performance is often blamed on optimisers, priors, or noise models when the underlying issue is simpler: the Jacobian is nearly singular. A condition-number and Fisher-information analysis cannot solve the inverse problem by itself, but it can prevent one from asking the data for information they do not contain.

5.6. Local versus global identifiability

The round-trip tests demonstrate very accurate local recovery, but global identifiability is a separate matter. The three-mode system is especially vulnerable because it is exactly determined. Local minima can occur, and a small residual does not necessarily imply the correct parameter set when the search is started far away. This is why the five-mode result matters more for practice than the machine-precision local recovery.

Whether the spectral map is globally injective — that is, whether the oblate shell inverse problem admits a unique solution over the full parameter domain — remains an open question connecting to the spectral geometry programme initiated by Kac [1] and the perturbation theory of Kato [10]. The eigenvalue perturbation results of Hald [18, 19] and the shape optimisation framework of Henrot [20] provide relevant tools, but a proof of global uniqueness for the present coupled fluid–shell system would require substantial additional analysis.

Our recommendation is therefore not based on elegance but on behaviour:

- use three modes only for local studies or controlled round-trip tests;
- use at least five flexural modes for robust inversion in realistic settings.

5.7. Limitations

Several limitations remain.

1. The Ritz forward model is a variational approximation, not a full spheroidal shell solution. It captures the leading eccentricity dependence but incurs $\sim 8\%$ error relative to exact shell theory. Whether the rank restoration persists in a converged finite-element model remains to be confirmed.
2. The analysis is local. We do not prove global injectivity of the map $(a, c, E) \mapsto \{f_n\}$.
3. Synthetic inversion uses the same model for data generation and inversion. This isolates identifiability from model-form error, but it means that the round-trip recovery (Section 4) is a verification of internal consistency, not a demonstration of real-world inversion capability.
4. The mode numbers are assumed known. In experiment, mode assignment may itself be uncertain.
5. Only three parameters are estimated. Allowing h, ν, or fluid properties to vary would enlarge the parameter space and may degrade conditioning.

5.8. Future directions

The natural next step is a full finite-element oblate shell model, which would eliminate the Ritz truncation error and allow convergence to the exact spheroidal solution. Shape-calculus tools [21] would allow Hadamard-type Fréchet derivatives of the eigenvalues with respect to the domain boundary, enabling a geometry-aware sensitivity analysis that goes beyond the parametric approach taken here. That would show whether the encouraging results reported here strengthen further when the geometry is represented exactly. The second priority is experimental validation, ideally first on phantoms of known geometry and stiffness and then on biological specimens. A Bayesian treatment, with priors on tissue properties and uncertainty in mode identification, would also be a useful extension.

The results suggest that geometric asymmetry plays a fundamental role in practical identifiability of coupled fluid–shell systems, a connection that merits further investigation using the tools of shape calculus and spectral perturbation theory [10, 20].

5.9. Classification of shell inverse problems

The two-channel expansion (27) and the oblate–prolate comparison together suggest a three-class taxonomy for the identifiability of spectral inverse problems on fluid-filled shells.

1. **Singular** (Jacobian rank-deficient). The equivalent-sphere model, which depends on (a, c) only through R_{eq}, has a formally infinite condition number by Proposition 1 (evaluated numerically as $\kappa \approx 1.37 \times 10^{10}$ with finite-difference step $\delta = 10^{-6}|\theta_j|$). This is the degenerate case: all frequency sensitivities to a and c are proportional, leaving one parameter combination unobservable.
2. **Regular bounded** ($\sigma_0 > 0$, κ finite and generally favourable). The oblate Ritz model has a positive curvature-channel floor $\sigma_0 = 0.01113$ (five modes), giving $\kappa_{\mathrm{floor}} \approx 269$ at the sphere and generally improving to $\kappa = 69.4$ at the canonical oblate operating point — a factor of 3.9. The improvement is not strictly monotonic: local non-monotonicity attributable to mode crossings in the Ritz eigenvalue branches produces isolated spikes in κ

at intermediate eccentricities. Because $\varepsilon_c = 1.48 > 1$, the curvature channel dominates everywhere. The inverse problem is well posed at all physically realisable eccentricities, though the practitioner should be aware that the condition number may vary non-smoothly along the oblate branch.

3. **Non-lifting** (κ high and essentially flat). For prolate geometries ($c > a$), the condition number remains in the range 580–730 with no systematic improvement as eccentricity increases. The inverse problem is not rank-deficient, but it is poorly conditioned and does not benefit from departure from the sphere.

The distinction between classes 2 and 3 is the central new finding. It replaces an earlier conjecture [3, Conjecture 1, first draft] that proposed a power-law divergence $\kappa \sim C\varepsilon^{-\alpha}$ as $\varepsilon \to 0$. That conjecture was abandoned because the Ritz model has $\sigma_0 > 0$ at the sphere: the condition number approaches a finite floor, not infinity. The two-channel expansion provides a richer and more accurate description.

5.10. Oblate–prolate asymmetry

The three-class taxonomy raises a natural question: *why* does oblate geometry improve identifiability while prolate geometry does not?

The answer lies in the spatial relationship between curvature concentration and modal displacement peaks — a mechanism we term *curvature–mode anti-correlation.*

For an oblate spheroid, the Gaussian curvature is maximal at the equator and minimal at the poles. The low-order Legendre modes $P_n(\cos\theta)$, by contrast, have their largest amplitudes near the poles (where $\cos\theta \to 1$). Different modes therefore sample distinct curvature environments: low-order modes weight the low-curvature polar cap, while higher-order modes concentrate energy in the high-curvature equatorial belt. This spatial diversity means that each mode's sensitivity to a and c differs, breaking the proportionality of the Jacobian columns and reducing κ.

For a prolate spheroid, the geometry reverses. Gaussian curvature is now maximal at the poles — exactly where all Legendre modes peak. Every mode therefore samples essentially the same high-curvature region with similar weighting. The Jacobian columns remain nearly proportional regardless of eccentricity, and κ stays high and flat.

The mechanism is constructive rather than symmetry-theoretic: it identifies the specific integrals $\int_0^1 P_n^2(\eta)\, G^{-k}(\eta)\, \mathrm{d}\eta$ whose n-dependence does or does not vary with eccentricity. In the oblate case, the factor $(\eta^2 - \frac{2}{3})$ in the perturbation integral changes sign on $[0, 1]$, and different P_n^2 sample the positive and negative regions differently. In the prolate case, the analogous factor does not produce sufficient n-dependent variation, because the modal peaks and curvature peaks are co-located.

5.11. Open questions

The two-channel expansion and three-class classification organise the numerical findings, but several questions remain open. We record the outstanding mathematical programme explicitly.

Let $\mathcal{N} = \{n_1, \ldots, n_N\}$ denote the measured flexural mode set, let $\varepsilon = \sqrt{1 - c^2/a^2}$ be the eccentricity, and let $\mathbf{J}_s(\varepsilon; \mathcal{N})$ denote the corresponding scaled Jacobian of the frequency–parameter map.

Open question 1: closed-form expansion coefficients.. For the Ritz model with fixed mode set $\mathcal{N}$, can the expansion coefficients $\sigma_0(\mathcal{N})$, $\lambda_1(\mathcal{N})$, and $\lambda_2(\mathcal{N})$ in (27) be expressed in closed form as functions of the curvature-weighted Legendre integrals $\int_0^1 P_n^2(\eta)\,(\eta^2 - \frac{2}{3})\,\mathrm{d}\eta$ and the canonical parameter set? A derivation would turn the numerically determined coefficients (39) into design formulae, giving the floor, slope, and crossover eccentricity as explicit functions of the mode set.

Open question 2: minimum modes for identifiability.. Consider the four-parameter material map with parameter vector

$$\boldsymbol{\vartheta} = (E, h, \rho_w, \rho_f)^\top, \tag{41}$$

at fixed geometry and fixed $(\nu, K_f, P_{\mathrm{iap}})$. The Ritz forward model introduces three independent dimensionless groups that govern the frequency spectrum (beyond ε itself): a stiffness ratio Eh/K_fR, a density ratio $\rho_w h/\rho_f R$, and a prestress ratio P_{iap}/E. If these three groups exhaust the independent parameter combinations (as the structure of equations (3)–(2) suggests but we do not formally prove), then three measured flexural modes suffice for local identifiability if and only if $\varepsilon > 0$. At $\varepsilon = 0$, the additional proportionality constraint of Proposition 1 removes one degree of freedom, and no finite mode set yields full column rank. The completeness of the three-group reduction has not been established rigorously; a formal proof would require verifying that no hidden similarity scaling further reduces the effective parameter dimension.

Remark 5 (Monotone improvement with mode count). *The numerical evidence of Table 2 shows that the Cramér–Rao bounds improve monotonically as the mode count $\mathcal{N}$ increases from 3 to 9, with diminishing marginal returns beyond $\mathcal{N} = 5$. This is an expected consequence of overdetermination: each additional mode adds an independent row to the Jacobian, reducing the covariance bound. However, we observe no clean power-law decay of κ with $\mathcal{N}$ at fixed ε — the condition number is non-monotone (Table 2), ranging from 69.4 at $\mathcal{N} = 5$ to 93.6 at $\mathcal{N} = 9$. We therefore record the improvement as an empirical observation rather than elevating it to a formal conjecture. The practical implication is that five modes capture the bulk of the available information for the present three-parameter problem.*

Open Question 1 is the most tantalising of these. The numerics already determine the expansion coefficients to several significant figures (39), but a closed-form derivation would give the floor, slope, and crossover as explicit functions of the mode set — turning a useful fit into an asymptotic theorem. That would be the crown jewel of the present identifiability programme; at the time of writing, it remains work in progress.

6. Conclusion

We have studied the practical identifiability of elastic shell parameters from resonant frequencies in the spirit of Kac's spectral question. For the oblate Ritz flexural model used here, the inverse problem based on the parameter triple (a, c, E) transitions smoothly from poorly conditioned at the near-spherical limit to well conditioned at the canonical oblate operating point. In the context of the Browntone programme, that makes Paper 8 less a side study than a foundation stone: it addresses the inverse question on which the diagnostic ambitions of Papers 1–7 depend.

The main findings are:

1. within the Ritz model, the condition number drops from a near-spherical floor value $\kappa_{\text{floor}} \approx 269$ (five modes) to $\kappa = 69.4$ (canonical oblate), purely through geometric eccentricity;
2. the equivalent-sphere model is algebraically rank deficient (formally $\kappa = \infty$; evaluated numerically as $\kappa \approx 1.37 \times 10^{10}$ due to the finite-difference step), illustrating how model-induced degeneracy dwarfs the geometric effect;
3. the smallest singular value admits a regular two-channel expansion $\sigma_{\min} = \sigma_0 + \lambda_1 \varepsilon^2 + O(\varepsilon^4)$ with curvature-channel floor $\sigma_0 = 0.01113$ and shape-channel coefficient $\lambda_1 = 0.00507$ (five modes); the crossover $\varepsilon_c = 1.48 > 1$ shows that the curvature channel dominates across the entire physical range, and the condition number saturates at $\kappa_{\text{floor}} \approx 269$ rather than diverging;
4. round-trip inversion recovers (a, c, E) to machine precision ($< 10^{-12}$), confirming internal consistency of the forward model and solver;
5. at 1% frequency noise (five modes), the Cramér–Rao bounds are approximately ±22.5% for a, ±6.0% for c, and ±30.8% for E, comparable to existing elastographic measurement precision;
6. the condition number remains in the range $\kappa \in [69, 94]$ across measured mode counts $\mathcal{N} = 3$–9, demonstrating robustness to measurement configuration (though this is not Ritz basis convergence evidence, since the trial-function order is held fixed);
7. three-mode inversion is vulnerable to local minima, whereas five-mode inversion is robust across the tested initial guesses;
8. a ripe-watermelon case remains well conditioned, with $\kappa \approx 128$.

These results suggest a practical answer to the question posed in the title. Within the present model, one can hear the shape of an oblate organ in the limited but useful sense of practical local identifiability. One cannot do so reliably at the spherical limit. That distinction is precisely what determines whether a measured resonance is merely interesting or diagnostically useful.

The programme-level implication is straightforward. If flexural resonances can be measured reproducibly, and if the relevant parameters are identifiable, then the spectrum becomes a non-invasive elastography tool. For abdominal resonance [3], this means that a measured modal set could in principle constrain effective cavity geometry and wall stiffness rather than serving only as a forward prediction. For the bladder problem [4], it means that fill-dependent resonance could, at least in principle, support inference of changing wall properties. For watermelon ripeness [5], it explains why a tap-tone can carry recoverable material information rather than folklore alone.

The clearest practical recommendation is equally straightforward: measure at least five low-order flexural resonances if the goal is parameter inversion rather than mere modal description. Fewer modes may suffice for local tests, but they do not provide the same robustness.

The main theoretical contribution is the three-class classification of shell inverse problems (Section 5.9): *singular* (equivalent-sphere, rank-deficient), *regular bounded* (oblate Ritz, κ finite and generally improving with eccentricity, though with local non-monotonicity from mode crossings in the Ritz eigenvalue branches), and *non-lifting* (prolate, κ high and flat). The critical finding is that not all shape departures improve identifiability: oblate geometry benefits

from curvature–mode anti-correlation, while prolate geometry does not, because curvature peaks and modal peaks are co-located at the poles. This asymmetry was not anticipated by the original power-law conjecture, which has been replaced by the richer two-channel picture.

The most interesting open mathematical target is the closed-form derivation of the expansion coefficients σ_0, λ_1, and λ_2 as explicit functions of the mode set and canonical parameters (Open Question 1, Section 5.11). The numerically determined values (39) already provide quantitative design rules, but a proof would turn them into a theorem connecting geometry, modal structure, and inverse conditioning in closed form. That would be the crown jewel. For the moment, the jewel remains in the workshop.

Future work should therefore proceed in two directions. First, the Ritz approximation should be replaced by a full finite-element oblate shell model to eliminate truncation error. Second, the inversion should be tested experimentally on phantoms and biological specimens. Together, those steps would move the Browntone programme from "the resonances can be predicted" to "the material parameters can be recovered", which is the more consequential claim.

References

[1] M. Kac, Can one hear the shape of a drum?, The American Mathematical Monthly 73 (4) (1966) 1–23. doi:10.2307/2313748.

[2] C. Gordon, D. L. Webb, S. Wolpert, One cannot hear the shape of a drum, Bulletin of the American Mathematical Society 27 (1) (1992) 134–138. doi:10.1090/S0273-0979-1992-00289-6.

[3] J. Mace, B. R. Mace, The brown note revisited: Infrasonic resonance of the human abdominal cavity, companion paper (Paper 1), submitted to Journal of Sound and Vibration (2026).

[4] J. Mace, B. R. Mace, Fill-dependent resonance of the human urinary bladder: Competing geometry and stiffening in a fluid-filled viscoelastic shell, companion paper (Paper 4), submitted to Journal of Sound and Vibration (2026).

[5] J. Mace, B. R. Mace, Can you hear the ripeness? non-destructive acoustic assessment of fruit maturity via equivalent-sphere shell resonance inversion, companion paper (Paper 7), submitted to Postharvest Biology and Technology (2026).

[6] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, J. Timmer, Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood, Bioinformatics 25 (15) (2009) 1923–1929. doi:10.1093/bioinformatics/btp358.

[7] F.-G. Wieland, A. L. Hauber, M. Rosenblatt, C. Tönsing, J. Timmer, On structural and practical identifiability, Current Opinion in Systems Biology 25 (2021) 60–69. doi:10.1016/j.coisb.2021.03.005.

[8] W. Soedel, Vibrations of Shells and Plates, 3rd Edition, Marcel Dekker, 2004. doi:10.1201/9780203026304.

[9] A. W. Leissa, Vibration of Shells, NASA, 1973, NASA SP-288.

[10] T. Kato, Perturbation Theory for Linear Operators, Vol. 132 of Grundlehren der mathematischen Wissenschaften, Springer, 1966. doi:10.1007/978-3-642-66282-9.

[11] G. W. Stewart, J.-g. Sun, Matrix Perturbation Theory, Academic Press, Boston, 1990.

[12] M. M. Doyley, Model-based elastography: A survey of approaches to the inverse elasticity problem, Physics in Medicine and Biology 57 (3) (2012) R35–R73. doi:10.1088/0031-9155/57/3/R35.

[13] R. G. Barr, G. Ferraioli, M. L. Palmeri, Z. D. Goodman, G. Garcia-Tsao, J. Rubin, B. Garra, R. P. Myers, S. R. Wilson, D. Rubens, D. Levine, Elastography assessment of liver fibrosis: Society of radiologists in ultrasound consensus conference statement, Radiology 276 (3) (2015) 845–861. doi:10.1148/radiol.2015150619.

[14] J. Sylvester, G. Uhlmann, A global uniqueness theorem for an inverse boundary value problem, Annals of Mathematics 125 (1) (1987) 153–169. doi:10.2307/1971291.

[15] G. M. L. Gladwell, Inverse problems in vibration, Applied Mechanics Reviews 57 (6) (2004) 401–422. doi:10.1115/1.1795979.

[16] G. M. L. Gladwell, Inverse Problems in Vibration, 2nd Edition, Vol. 119 of Solid Mechanics and Its Applications, Springer, 2005. doi:10.1007/1-4020-2721-4.

[17] J. Aernouts, I. Couckuyt, K. Crombecq, T. Dhaene, Inverse estimation of material properties from dynamic measurements of a tympanic membrane model, Journal of the Acoustical Society of America 131 (6) (2012) 4648–4660.

[18] O. H. Hald, The inverse Sturm–Liouville problem and the Rayleigh–Ritz method, Mathematics of Computation 32 (143) (1978) 687–705. doi:10.2307/2006482.

[19] O. H. Hald, The inverse Sturm–Liouville problem and the Rayleigh–Ritz method, Communications on Pure and Applied Mathematics 37 (4) (1984) 539–577. doi:10.1002/cpa.3160370406.

[20] A. Henrot, Extremum Problems for Eigenvalues of Elliptic Operators, Frontiers in Mathematics, Birkhäuser, 2006. doi:10.1007/3-7643-7706-2.

[21] M. C. Delfour, J.-P. Zolésio, Shapes and Geometries: Metrics, Analysis, Differential Calculus, and Optimization, 2nd Edition, Vol. 22 of Advances in Design and Control, SIAM, Philadelphia, 2011. doi:10.1137/1.9780898719826.

Data availability

All code and data required to reproduce the results are available in the accompanying repository at https://github.com/JonathanMace/brownnote.

CRediT authorship contribution statement

Jonathan Mace: Conceptualisation, Methodology, Software, Writing – Original Draft. **Brian R. Mace**: Supervision, Writing – Review & Editing.

Acknowledgements

The authors gratefully acknowledge the support of Microsoft Research and the University of Auckland Acoustics Research Centre.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.